

Bosch Seamless Panel Usability Test

Usability Test Report

Generated: 24/03/2026

Observation sheet for usability testing of the Bosch Seamless Touch Panel water heater controller. Covers simple identification/control tasks (S1-S7) and complex multi-step tasks (C1-C7).

Summary

Total Sessions	4
Completed	4
In Progress	0
Planned	0

Session 1: Participant-fdab97

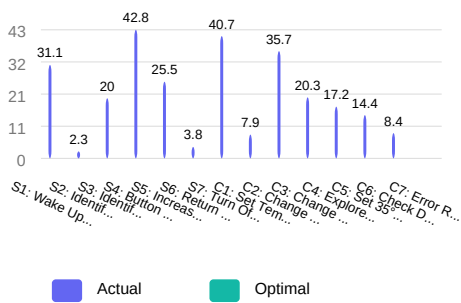
Participant	Participant-fdab97
Age	21
Gender	female
Tech Proficiency	low
Evaluator	Hugo Correia
Status	completed
Started	24/03/2026, 18:19:25
Completed	24/03/2026, 18:32:37

Task Results

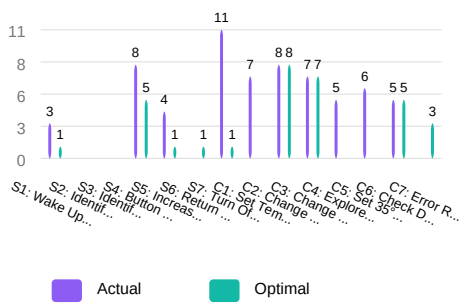
Task	Comp.	Time (s)	Actions	Optimal	Errors	Hesit.	SEQ
S1: Wake Up the Panel	failure	31.1	3	1	3	0	1
S2: Identify Current Time	success	2.3	0	0	0	0	7
S3: Identify All Buttons	partial	20.0	0	—	0	0	6
S4: Button Responsibilities	success	42.8	8	5	1	0	5
S5: Increase Temp +1°C	success	25.5	4	1	1	0	4
S6: Return to Previous	success	3.8	0	1	0	0	7
S7: Turn Off Panel	success	40.7	11	1	10	0	5
C1: Set Temp to 40°C	success	7.9	7	—	0	0	6
C2: Change Clock to 08:30	success	35.7	8	8	0	0	6
C3: Change Language	partial	20.3	7	7	3	0	5
C4: Explore Menu Structure	success	17.2	5	—	0	0	6
C5: Set 35°C + Return	success	14.4	6	—	1	0	5
C6: Check Device Info	partial	8.4	5	5	2	0	5
C7: Error Recovery	failure	0.0	0	3	0	0	1

Session Performance Charts

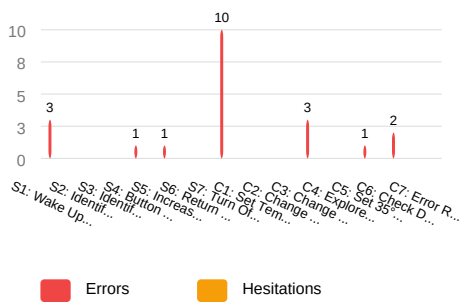
Time vs Optimal (seconds)



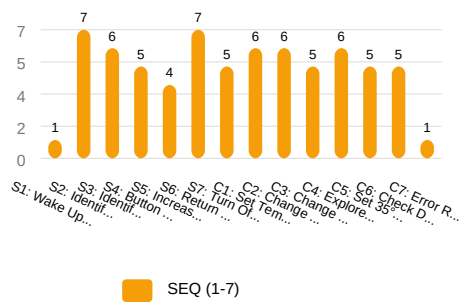
Actions vs Optimal



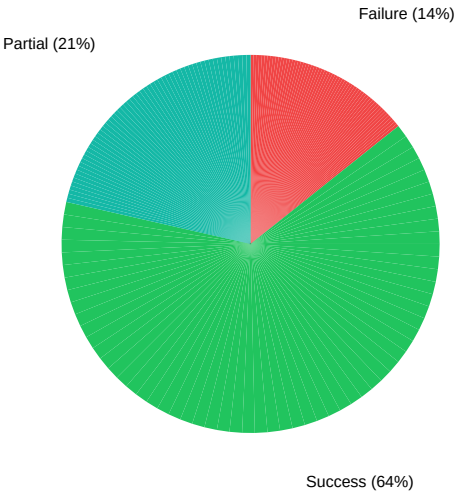
Errors & Hesitations per Task



SEQ Ratings per Task



Task Completion Status



Error Log

Task	Time (s)	Description
S1: Wake Up the Panel	28.6	—
S1: Wake Up the Panel	28.3	—
S1: Wake Up the Panel	28.8	—
S4: Button Responsibilities	23.3	—
S5: Increase Temp +1°C	4.8	—
S7: Turn Off Panel	3.9	—
S7: Turn Off Panel	7.9	—
S7: Turn Off Panel	10.6	—
S7: Turn Off Panel	13.4	—
S7: Turn Off Panel	16.4	—
S7: Turn Off Panel	21	—
S7: Turn Off Panel	23.9	—
S7: Turn Off Panel	30	—
S7: Turn Off Panel	38	—
S7: Turn Off Panel	39.8	—
C3: Change Language	5.4	—
C3: Change Language	5.6	—
C3: Change Language	10.2	—
C5: Set 35°C + Return	8.5	—
C6: Check Device Info	8.4	—
C6: Check Device Info	8.4	—

Interview Answers

Q: What was the most frustrating part of using this panel?

a quantidade de erros e problemas nas funcionalidades. o facto para mudar a data era necessario ir ao menu mas para mudar a temperatura nao apesar de "Temperatura" existir no menu

Q: Was there anything the panel did that surprised you or that you didn't expect?

a quantidade de erros e falta de sensibilidade do botao de ligar e a multi-funcionalidade dos botoes

Q: If you could change one thing about this interface, what would it be?

os botoes podiam ter cores diferentes que ajudassem a entender a sua funcionalidade ou desaparecer e aparecer caso nao podessem ser pressionados, por exemplo, quando o painel esta a aquecer

Q: How would you compare this to a traditional physical-button water heater controller?

o tradicional e mais intuitivo e menos incongruente

SUS Questionnaire — Score: 25.0 (Poor)

#	Question	Score
1	I think that I would like to use this system frequently.	2/5
2	I found the system unnecessarily complex.	4/5
3	I thought the system was easy to use.	3/5
4	I think that I would need the support of a technical person to be able to use this system.	3/5
5	I found the various functions in this system were well integrated.	1/5
6	I thought there was too much inconsistency in this system.	5/5
7	I would imagine that most people would learn to use this system very quickly.	2/5
8	I found the system very cumbersome to use.	4/5
9	I felt very confident using the system.	1/5
10	I needed to learn a lot of things before I could get going with this system.	3/5

Session 2: Participant-e4e581

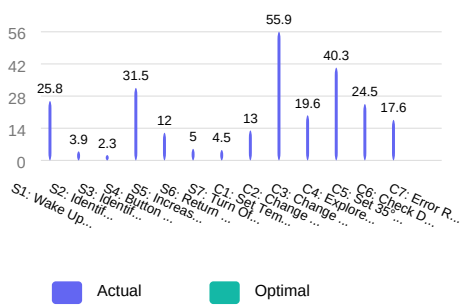
Participant	Participant-e4e581
Age	22
Gender	female
Tech Proficiency	low
Evaluator	Hugo Correia
Status	completed
Started	24/03/2026, 19:53:45
Completed	24/03/2026, 20:09:06

Task Results

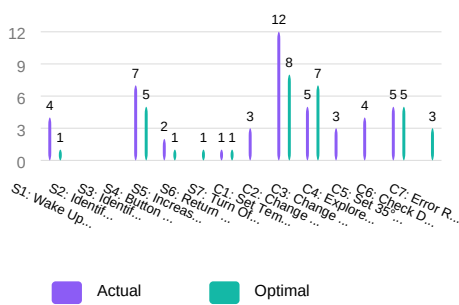
Task	Comp.	Time (s)	Actions	Optimal	Errors	Hesit.	SEQ
S1: Wake Up the Panel	failure	25.8	4	1	1	0	1
S2: Identify Current Time	success	3.9	0	0	0	0	7
S3: Identify All Buttons	success	2.3	0	—	0	0	7
S4: Button Responsibilities	partial	31.5	7	5	0	0	6
S5: Increase Temp +1°C	success	12.0	2	1	0	0	6
S6: Return to Previous	success	5.0	0	1	0	0	6
S7: Turn Off Panel	success	4.5	1	1	0	0	7
C1: Set Temp to 40°C	success	13.0	3	—	2	0	3
C2: Change Clock to 08:30	success	55.9	12	8	7	0	6
C3: Change Language	partial	19.6	5	7	2	0	3
C4: Explore Menu Structure	success	40.3	3	—	5	0	6
C5: Set 35°C + Return	success	24.5	4	—	2	0	5
C6: Check Device Info	partial	17.6	5	5	0	0	6
C7: Error Recovery	failure	0.0	0	3	0	0	1

Session Performance Charts

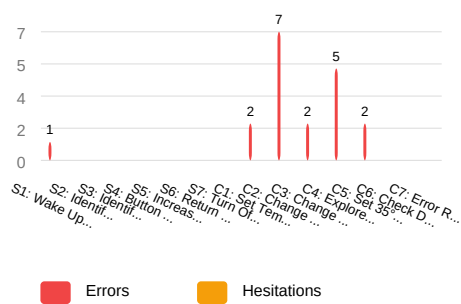
Time vs Optimal (seconds)



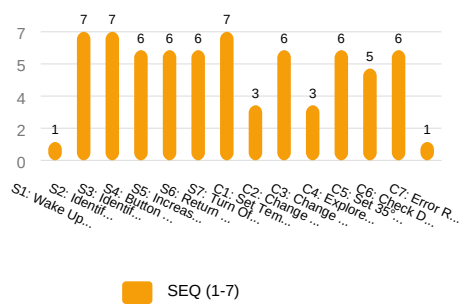
Actions vs Optimal



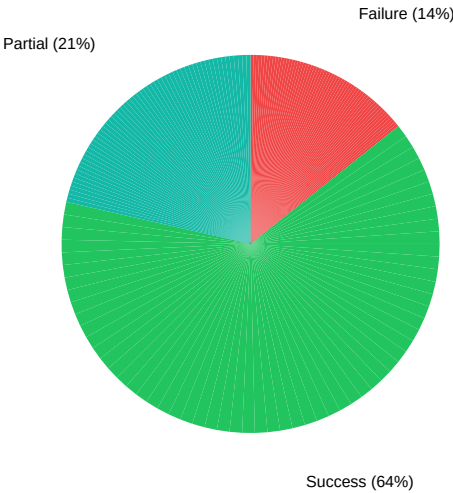
Errors & Hesitations per Task



SEQ Ratings per Task



Task Completion Status



Error Log

Task	Time (s)	Description
S1: Wake Up the Panel	10	—
C1: Set Temp to 40°C	8.1	—
C1: Set Temp to 40°C	8.3	—
C2: Change Clock to 08:30	6.8	—
C2: Change Clock to 08:30	7.1	—
C2: Change Clock to 08:30	9.6	—
C2: Change Clock to 08:30	10.9	—
C2: Change Clock to 08:30	13.3	—
C2: Change Clock to 08:30	20.5	—
C2: Change Clock to 08:30	38.7	—
C3: Change Language	4.3	—
C3: Change Language	13	—
C4: Explore Menu Structure	4.5	—
C4: Explore Menu Structure	33.7	—
C4: Explore Menu Structure	34	—
C4: Explore Menu Structure	34.5	—
C4: Explore Menu Structure	37.4	—
C5: Set 35°C + Return	8.6	—
C5: Set 35°C + Return	9	—

Interview Answers

Q: What was the most frustrating part of using this panel?

O botão de retroceder não funcionar quando está no “menu” onde apresenta as horas. Não é intuitivo ter de seleccionar os botões cima e baixo para ir para sair de onde mostra as horas.

Q: Was there anything the panel did that surprised you or that you didn't expect?

O que me surpreendeu mais foi o facto de não ser mesmo nada intuitivo a ligar.

Q: If you could change one thing about this interface, what would it be?

Botão de ligar, o facto do painel estar bastante desfocado (mau visualmente e pode até impossibilitar algumas pessoas com problemas de visão de conseguir ler), botão de retroceder, adicionar informações ao menu.

Q: How would you compare this to a traditional physical-button water heater controller?

Pouco intuitivo

SUS Questionnaire — Score: 40.0 (Poor)

#	Question	Score
1	I think that I would like to use this system frequently.	1/5
2	I found the system unnecessarily complex.	1/5
3	I thought the system was easy to use.	3/5
4	I think that I would need the support of a technical person to be able to use this system.	3/5
5	I found the various functions in this system were well integrated.	2/5
6	I thought there was too much inconsistency in this system.	5/5

#	Question	Score
7	I would imagine that most people would learn to use this system very quickly.	1/5
8	I found the system very cumbersome to use.	4/5
9	I felt very confident using the system.	3/5
10	I needed to learn a lot of things before I could get going with this system.	1/5

Session 3: Participant-fd8b16

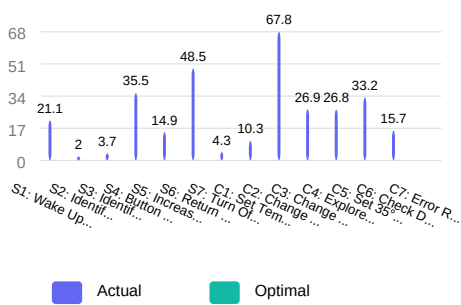
Participant	Participant-fd8b16
Age	20
Gender	male
Tech Proficiency	high
Evaluator	Hugo Correia
Status	completed
Started	24/03/2026, 20:23:48
Completed	24/03/2026, 20:38:21

Task Results

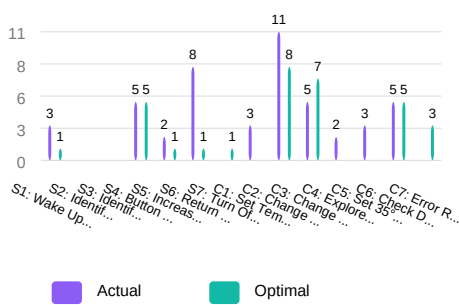
Task	Comp.	Time (s)	Actions	Optimal	Errors	Hesit.	SEQ
S1: Wake Up the Panel	partial	21.1	3	1	0	0	4
S2: Identify Current Time	success	2.0	0	0	0	0	7
S3: Identify All Buttons	success	3.7	0	—	0	0	7
S4: Button Responsibilities	success	35.5	5	5	2	0	6
S5: Increase Temp +1°C	success	14.9	2	1	0	0	7
S6: Return to Previous	partial	48.5	8	1	3	2	4
S7: Turn Off Panel	success	4.3	0	1	0	0	6
C1: Set Temp to 40°C	success	10.3	3	—	0	0	7
C2: Change Clock to 08:30	success	67.8	11	8	5	0	5
C3: Change Language	partial	26.9	5	7	4	0	5
C4: Explore Menu Structure	success	26.8	2	—	6	0	5
C5: Set 35°C + Return	success	33.2	3	—	3	0	6
C6: Check Device Info	partial	15.7	5	5	2	0	6
C7: Error Recovery	failure	0.0	0	3	0	0	1

Session Performance Charts

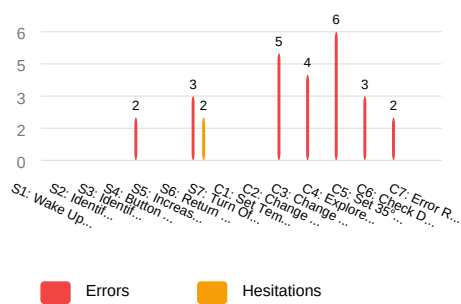
Time vs Optimal (seconds)



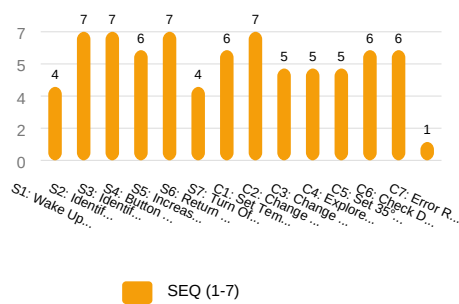
Actions vs Optimal



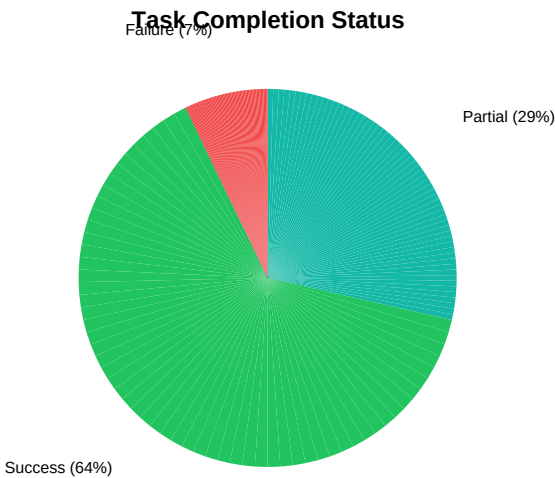
Errors & Hesitations per Task



SEQ Ratings per Task



Task Completion Status



Error Log

Task	Time (s)	Description
S4: Button Responsibilities	13.7	—
S4: Button Responsibilities	26	—
S6: Return to Previous	4.2	—
S6: Return to Previous	4.3	—
S6: Return to Previous	4.6	—
C2: Change Clock to 08:30	14.1	—
C2: Change Clock to 08:30	17.2	—
C2: Change Clock to 08:30	35.1	—
C2: Change Clock to 08:30	35.2	—
C2: Change Clock to 08:30	38.3	—
C3: Change Language	7.2	—
C3: Change Language	7.4	—
C3: Change Language	11	—
C3: Change Language	12.6	—
C4: Explore Menu Structure	11	—
C4: Explore Menu Structure	10.9	—
C4: Explore Menu Structure	22.9	—
C4: Explore Menu Structure	23.1	—
C4: Explore Menu Structure	25.6	—
C4: Explore Menu Structure	25.8	—
C5: Set 35°C + Return	5.7	—
C5: Set 35°C + Return	5.9	—
C5: Set 35°C + Return	15	—
C6: Check Device Info	2.5	—
C6: Check Device Info	4	—

Hesitation Notes

Task	Time (s)	Note
S6: Return to Previous	29.5	—
S6: Return to Previous	30.6	—

Interview Answers

Q: What was the most frustrating part of using this panel?

Very frustrating at first because the discovery is not intuitive.

Q: Was there anything the panel did that surprised you or that you didn't expect?

Going back to the menu after cooling.

Q: If you could change one thing about this interface, what would it be?

More info about user interactions.

Q: How would you compare this to a traditional physical-button water heater controller?

Bad

SUS Questionnaire — Score: 17.5 (Awful)

#	Question	Score
1	I think that I would like to use this system frequently.	1/5
2	I found the system unnecessarily complex.	5/5
3	I thought the system was easy to use.	1/5
4	I think that I would need the support of a technical person to be able to use this system.	2/5
5	I found the various functions in this system were well integrated.	2/5
6	I thought there was too much inconsistency in this system.	5/5
7	I would imagine that most people would learn to use this system very quickly.	1/5
8	I found the system very cumbersome to use.	5/5
9	I felt very confident using the system.	2/5
10	I needed to learn a lot of things before I could get going with this system.	3/5

Session 4: Participant-189fe9

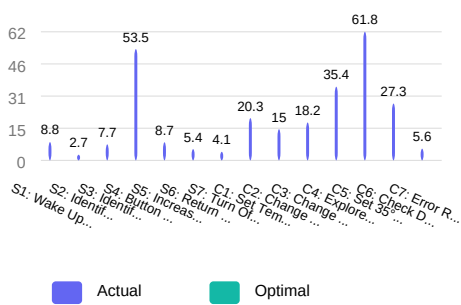
Participant	Participant-189fe9
Age	23
Gender	male
Tech Proficiency	low
Evaluator	Hugo Correia
Status	completed
Started	24/03/2026, 17:39:43
Completed	24/03/2026, 17:55:05

Task Results

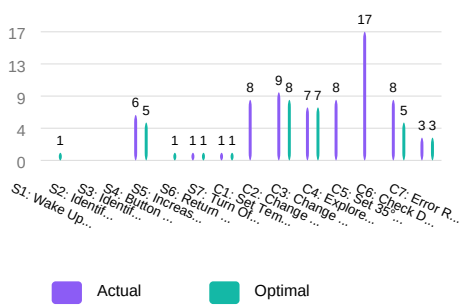
Task	Comp.	Time (s)	Actions	Optimal	Errors	Hesit.	SEQ
S1: Wake Up the Panel	failure	8.8	0	1	0	0	1
S2: Identify Current Time	success	2.7	0	0	0	0	7
S3: Identify All Buttons	success	7.7	0	—	0	0	6
S4: Button Responsibilities	partial	53.5	6	5	1	0	4
S5: Increase Temp +1°C	success	8.7	0	1	0	0	7
S6: Return to Previous	success	5.4	1	1	0	0	7
S7: Turn Off Panel	success	4.1	1	1	0	0	7
C1: Set Temp to 40°C	success	20.3	8	—	0	0	6
C2: Change Clock to 08:30	success	15.0	9	8	0	0	7
C3: Change Language	success	18.2	7	7	0	0	5
C4: Explore Menu Structure	success	35.4	8	—	0	2	6
C5: Set 35°C + Return	success	61.8	17	—	3	0	4
C6: Check Device Info	success	27.3	8	5	2	0	4
C7: Error Recovery	success	5.6	3	3	0	0	7

Session Performance Charts

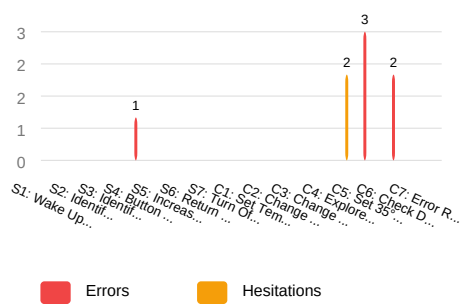
Time vs Optimal (seconds)



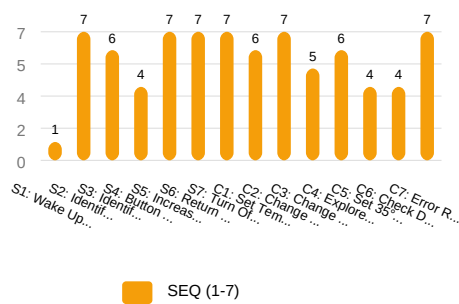
Actions vs Optimal



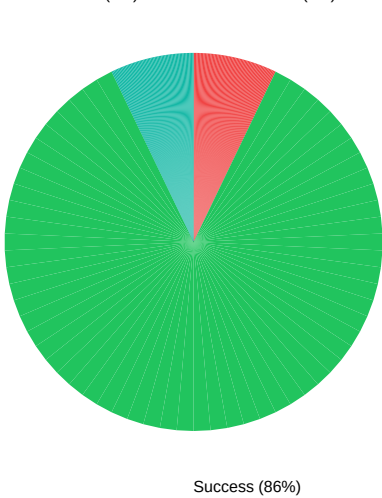
Errors & Hesitations per Task



SEQ Ratings per Task



Task Completion Status



Hesitation Notes

Task	Time (s)	Note
C4: Explore Menu Structure	24.4	—
C4: Explore Menu Structure	24.9	—

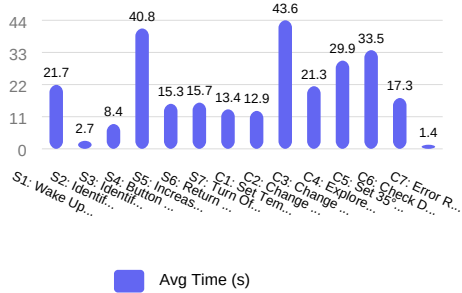
SUS Questionnaire — Score: 30.0 (Poor)

#	Question	Score
1	I think that I would like to use this system frequently.	2/5
2	I found the system unnecessarily complex.	4/5
3	I thought the system was easy to use.	2/5
4	I think that I would need the support of a technical person to be able to use this system.	3/5
5	I found the various functions in this system were well integrated.	2/5
6	I thought there was too much inconsistency in this system.	4/5
7	I would imagine that most people would learn to use this system very quickly.	2/5
8	I found the system very cumbersome to use.	4/5
9	I felt very confident using the system.	3/5
10	I needed to learn a lot of things before I could get going with this system.	4/5

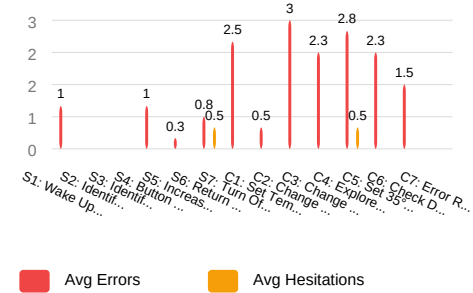
Overall Analysis

Aggregated data from 4 completed sessions

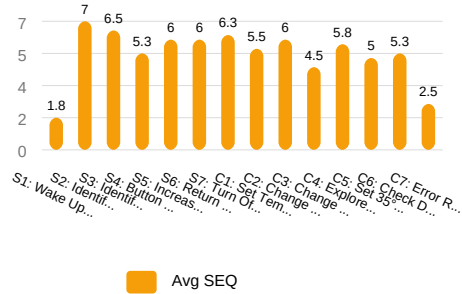
Average Time per Task (s)



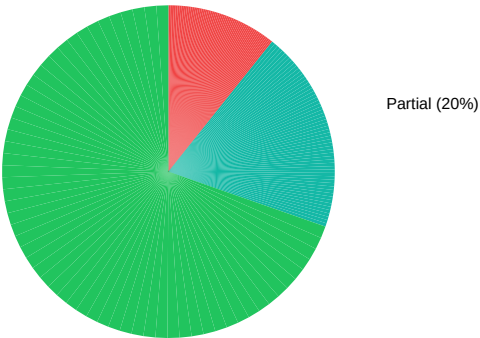
Avg Errors & Hesitations per Task



Average SEQ Rating per Task



Overall Task Completion



SUS Score Summary

Participant	SUS Score	Rating
Participant-fdab97	25.0	Poor
Participant-e4e581	40.0	Poor
Participant-fd8b16	17.5	Awful
Participant-189fe9	30.0	Poor
Average	28.1	Poor

SUS Scores Comparison

